We have successfully “solved” the 1D Ising model. Let’s add a bit more complexity by moving on to a 2D Ising model. In addition to addressing some of the simplifications we’ve made, we’ll introduce a new approach to neighborhood Q-learning called *mean field Q-learning*.

9.4. Mean field Q-learning and the 2D Ising model

You just saw how a neighborhood Q-learning approach is able to solve the 1D Ising model fairly rapidly. This is because, rather than using the full joint-action space that would have been a 2^{20} = 1,048,576 element joint-action vector, which is intractable, we just used each agent’s left and right neighbors. That reduced the size down to a 2^2 = 4 element joint-action vector, which is very manageable.

In a 2D grid, if we want to do the same thing and just get the joint-action space of an agent’s immediate neighbors, there are 8 neighbors, so the joint-action space is a 2^8 = 256 element vector. Computing with a 256 element vector is definitely doable, but doing it for say 400 agents in a 20 × 20 grid will start to get costly. If we wanted to use a 3D Ising model, the number of immediate neighbors would be 26 and the joint-action space is 2^{26} = 67,108,864; now we’re into intractable territory again.

As you can see, the neighborhood approach is much better than using the full joint-action space, but with more complex environments, even the joint-action space of immediate neighbors is too large when the number of neighbors is large. We need to make an even bigger simplifying approximation. Remember, the reason why the neighborhood approach works in the Ising model is because an electron’s spin is most affected by the magnetic field of its nearest neighbors. The magnetic field strength decreases proportionally to the square of the distance from the field source, so it is reasonable to ignore distant electrons.

We can make another approximation by noting that when two magnets are brought together, the resulting field is a kind of sum of these two magnets (figure 9.18). We can replace the knowledge of there being two separate